

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech (EE) Examination

March-April 2018

EE414 POWER SYSTEM OPERATIONS**Date: 06.04.2018, Friday****Time: 10.00 a.m. To 01.00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1** List out different operating states of power system. Explain each of them in detail. **[07]**
- Q – 2.(a)** Explain how the state estimation is used in power system. **[07]**
- (b)** What is the role of load management in demand side management? How is load control techniques classified? Discuss each of the techniques. **[07]**

OR

- Q – 2.(a)** Estimate two random variables and estimation error by weighted least square estimation method for a given measurement vector Z, H and Weight(W) as follows:- **[07]**

$$Z = \begin{bmatrix} 0.50 \\ 0.45 \\ 0.51 \end{bmatrix}, H = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \text{ and } W = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.5 & 0 \\ 0 & 0 & 0.1 \end{bmatrix}$$

- (b)** Write short notes on types of demand side management measures and programs. **[07]**
- Q - 3** **Answer any two.** **[14]**
- (a)** Give definition of the following terms: - (1) Man Machine Interface, (2) Modem, (3) LAN, (4) WAN, (5) Protocol (6) Remote terminal units, (7) Router.
- (b)** Discuss the following in relation with power system security evaluation
- (i) System monitoring
 - (ii) Contingency analysis
 - (iii) Corrective action analysis.
- (c)** What is the purpose of contingency analysis? Draw the flow chart of contingency analysis.

SECTION – II

- Q - 4** Which are the different entities in deregulated environment? Explain each of them in detail. **[07]**
- Q – 5.(a)** A circuit is formed by 4 components A, 3 component B and 3 components C all **[05]**

connected in series. The failure rates of component A, B, C are 4×10^{-3} , 5×10^{-3} and 6×10^{-3} failures/hour. Find (a) Mean time to failure (b) System reliability for an operating time of 12 hours. Assume exponential distribution.

OR

Q – 5.(a) A radial system consists of a transformer and a feeder. The feeder is 15 km long. [05]
The failure rates for transformer and feeder are 1.8 failure/year and 0.7 failures/km/year respectively. The repair times for transformer and feeder are 10 hours and 5 hours per outage. Find (a) Failure rate of system (b) total outage time per year (c) down time per outage.

Q – 5.(b) Two units of the system have the following cost curves: [04]

$$F(Pg_1) = 0.05Pg_1^2 + 22Pg_1 + 120 \text{ Rs/h}$$

$$F(Pg_2) = 0.06Pg_2^2 + 16Pg_2 + 120 \text{ Rs/h}$$

Where Pg in MW. Both units operate at all times and the maximum and minimum loads on each unit are 100 MW and 20 MW respectively. Determine the economic operating schedule of the plants for loads of 120 MW and 180 MW, neglecting the transmission line losses.

Q – 5.(c) Draw typical bath tub curve. Explain the three regions of the curve. [05]

OR

Q – 5.(c) A system consists of 4 components having reliabilities of 0.90, 0.94, 0.96 and 0.98. [05]
All the components are in series. Find the reliability of the system.

Q – 6.(a) Explain the double Auction method in restructured power system. [07]

Q – 6.(b) What do you mean by deregulation? Explain structure of deregulated industry. [07]

OR

Q – 6.(a) What are reasons for restructuring? Explain each of them in detail. [07]

Q – 6.(b) Write down the short notes on Congestion management. [07]
